

NAM T. THAN

PhD Candidate • Scientist • Engineer • Inventor

CURRICULUM VITAE

07/16/2026

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Email: thann2@ccf.org | **Publications:** [ORCID](#); [PubMed](#); [Google Scholar](#) | **Links:** [LinkedIn](#)

EDUCATION

- 08/2020 – Present **PhD** in Biomedical Engineering, The University of Texas at Austin, Austin, TX
Advisor: Hyun Jung Kim, PhD
- 01/2018 – 08/2020 **MS** in Biomedical Engineering, San Jose State University, San Jose, CA
Advisors: Melinda Simon, PhD (SJSU) & Aminat Adebisi, PhD (IBM Research)
- 06/2012 – 06/2016 **BS** in Biochemistry and Molecular Biology, University of California, Davis, Davis, CA

EXPERIENCES

- 07/2022 – Present **Predoctoral Research Fellow**, Department of Inflammation and Immunity, Cleveland Clinic Research, Cleveland, OH
- 09/2020 – 07/2022 **Graduate Research Assistant**, Department of Biomedical Engineering, UT Austin, Austin, TX
- 05/2018 – 08/2020 **Research Intern**, IBM Almaden Research Center, San Jose, CA
- 08/2018 – 05/2020 **Instructional Student Assistant**, Department of Biomedical Engineering, San Jose State University, San Jose, CA
- 10/2016 – 09/2017 **Formulation Lab Technician III**, Teknova, Hollister, CA

HONORS AND AWARDS

- 07/2026 – 12/2026 **Engineering Foundation Endowed Graduate Presidential Scholarship**, Cockrell School of Engineering, The University of Texas at Austin, Austin, TX
- 12/2024 **Mary B. Stark Travel Award**, Cleveland Clinic Research, Cleveland, OH
- 12/2023 **Mary B. Stark Travel Award**, Cleveland Clinic Research, Cleveland, OH
- 08/2023 – 08/2024 **Dean's Prestigious Fellowship Supplement**, College of Graduate Studies, The University of Texas at Austin, Austin, TX
- 06/2023 – 06/2024 **Gut Microbiome, Yogurt and Probiotics Fellowship Grant**, Danone North America, USA
- 05/2023 **Poster of Distinction**, Digestive Disease Week (2023; Chicago, IL)
- 04/2023 **William Lower Award**, Cleveland Clinic, Cleveland, OH
- 11/2020 **Professional Development Award**, College of Graduate Studies, The University of Texas at Austin, Austin, TX
- 10/2020 **Graduate Student Virtual Event Travel Awards**, College of Graduate Studies, San Jose State University, San Jose, CA
- 09/2020 – 09/2021 **Engineering Foundation Endowed Graduate Fellowship**, Cockrell School of Engineering, The University of Texas at Austin, Austin, TX
- 03/2012 **Einstein Award Winner**, The Synopsys Silicon Valley Science and Technology Outreach Foundation, San Jose, CA

PUBLICATIONS

- Jaeseung Youn*, Hyunji Kim*, **Nam Than**, Dohui Kim, Hyun Jung Kim, and Dong Sung Kim. [Nanofiber-supported decellularized matrix membrane creates a biomimetic biochemical and biophysical niche for intestinal organoid epithelium](#). *Adv. Healthc. Mater.*, **2026**, e01510. (*Equally contributed)
- Nam Than**, Tadanori Mammoto, Donald E. Ingber, and Hyun Jung Kim. [Probiotic Intervention Mitigates Radiation-Induced Intestinal Injury by Alleviating Oxidative Stress in a Human Gut-on-a-chip](#). *Sci. Rep.*, **2026**. (In Press)
- Nam Than** and Hyun Jung Kim. [Advanced Fabrication protocol of an Elastic Porous Membrane for Organ-on-a-chip Applications](#). *SLAS Technol.*, **2026**, 39, 100431.
- Soyoun Min, **Nam Than**, Yong Cheol Shin, Elif G. Ertugral, Chandrasekhar R. Kothapalli, Olumuyiwa Awoniyi, and Hyun Jung Kim. [Mechanostimulatory cues determine intestinal fibroblast fate and profibrotic remodeling in a physiodynamic human gut-on-a-chip](#). *Adv. Sci.*, **2026**, e16040.
- Yong Cheol Shin, **Nam Than**, Soojin Park, and Hyun Jung Kim. [Bioengineered human gut-on-a-chip for advancing non-clinical pharmaco-toxicology](#). *Expert Opin. Drug Metab. Toxicol.*, **2024**, 20(7), 593-606.
- Yong Cheol Shin*, **Nam Than***, Soyoun Min, Woojung Shin*, and Hyun Jung Kim*. [Modelling Host-Microbiome Interactions in Organs-on-chips](#). *Nat. Rev. Bioeng.*, **2023**, 2, 175–191. (*Equally contributed; *Co-Correspondence)

7. Soyoun Min*, **Nam Than***, Yong Cheol Shin, Grace Hu, Woojung Shin, Yoko M. Ambrosini, and Hyun Jung Kim. [Live probiotic bacteria administered in a pathomimetic Leaky Gut Chip ameliorate impaired epithelial barrier and mucosal inflammation](#). *Sci. Rep.*, **2023**, 12, 22641. (*Equally contributed)
8. **Nam Than***, Zamri Chik*, Amy Bowers, Luisa Bozano, and Aminat Adebisi. [Quantitation of ethanol in UTI assay for volatile organic compound detection by electronic nose using the validated headspace GC-MS method](#). *PloS ONE*, **2022**, 17(10), e0275517. (*Equally contributed)

CONFERENCE AND SEMINAR PRESENTATIONS

1. Soyoun Min*, **Nam Than**, Yong Cheol Shin, Olumuyiwa Awoniyi, and Hyun Jung Kim. Epithelial Barrier Dysfunction Drives Fluid Shear-induced Mechanoadaptive Fibroblasts and Profibrotic Remodeling in a Physiodynamic Human Gut-on-a-chip. *Digestive Disease Week (DDW)*, **2026**, Chicago, IL, USA (*Presenter; Poster)
2. **Nam Than**, Soyoun Min, Woojung Shin, and Hyun Jung Kim*. Modeling Microbiome-Mediated Therapeutics in a Patient-Derived IBD-on-a-Chip. *Keystone Symposia: Human Microbiome: From Models and Mechanism to Medicine*, **2026**, Banff, AB, Canada. (*Presenter; Poster)
3. **Nam Than***, Woojung Shin, Yoko M. Ambrosini, Soyoun Min, Yong Cheol Shin, and Hyun Jung Kim. FMT-qualified microbiota mitigates epithelial oxidative stress and inflammation in a patient-derived Crohn's disease-on-a-chip. *Cold Spring Harbor Laboratory Meetings: Bioengineered Tissue Systems & Models*, **2025**, Cold Spring Harbor, New York, USA. (*Presenter; Poster)
4. **Nam Than***, Soyoun Min, Yong Cheol Shin, and Hyun Jung Kim. Gut Microbiota Directly Controls Epithelial Oxidative Stress and Inflammatory Responses in a Patient-specific Crohn's Disease-on-a-chip. *Gastrointestinal Tract XXI: Life, Death, and Disease*, **2024**, Banff, AB, Canada. (*Speaker; Flash presentation on Sep 10, 2024)
5. **Nam Than***, Yong Cheol Shin, Soyoun Min, Woojung Shin, and Hyun Jung Kim. Modeling host-microbiome interactions in organ-on-a-chip platforms. *Inflammation and Immunity Trainee Seminar Series*, **2024**, Cleveland Clinic, Cleveland, OH, USA (*Speaker; Podium presentation on Apr 4, 2024)
6. **Nam Than***, Woojung Shin, Yoko M. Ambrosini, Soyoun Min, Yong Cheol Shin, and Hyun Jung Kim. Gut Microbiota Directly Controls Epithelial Oxidative Stress and Inflammatory Responses in a Patient-specific Crohn's Disease-on-a-chip. *Second Annual Microbiome Workshop*, **2024**, Case Comprehensive Cancer Center, Case Western Reserve University, OH, USA (*Speaker; Podium presentation on Feb 9, 2024)
7. **Nam Than***, Soyoun Min*, Yong Cheol Shin, Grace Hu, Woojung Shin, Yoko Ambrosini, and Hyun Jung Kim. Regenerative Effects of Live Probiotics on Mucosal Barrier and Inflammation in a Microengineered Pathomimetic Leaky Gut Chip. *Biomedical Engineering Society Annual Meeting*, **2023**, Seattle, WA, USA. (*Equally contributed; *Speaker; Podium presentation on Oct 12, 2023)
8. **Nam Than***, Woojung Shin*, Dennis Wylie, Yoko M. Ambrosini, Soyoun Min, Yong Cheol Shin, Hong Koh, Linda Annie Feagins, and Hyun Jung Kim. Gut Microbiota Directly Controls Epithelial Oxidative Stress and Inflammatory Responses in a Patient-specific Crohn's Disease-on-a-chip. *Digestive Disease Week (DDW)*, **2023**, Chicago, IL, USA (*Equally contributed; Poster of Distinction)
9. Yong Cheol Shin*, **Nam Than**, Soyoun Min, and Hyun Jung Kim. Piezo1-mediated Mechanostimulatory Regulation on Intestinal Epithelial Regeneration in a Physiodynamic Human Gut-on-a-chip. *Digestive Disease Week (DDW)*, **2023**, Chicago, IL, USA (*Presenter; Poster)
10. Soyoun Min*, Yong Cheol Shin, Alexander Wu, **Nam Than**, and Hyun Jung Kim. Immunotherapeutic Anti-Tumor Effect of Engineered T Cells in a Patient-specific Colorectal Cancer-on-a-chip. *Digestive Disease Week (DDW)*, **2023**, Chicago, IL, USA (*Presenter; Poster)
11. **Nam Than***, Woojung Shin, Yoko Ambrosini, Soyoun Min, Yong Cheol Shin, and Hyun Jung Kim. Gut Microbiota Orchestrates Pathological Signature in a Patient-specific Crohn's Disease-on-a-chip. *Inflammation and Immunity Trainee Seminar Series*, **2023**, Cleveland Clinic, Cleveland, OH, USA (*Speaker; Podium presentation on Mar 29, 2023)
12. Soyoun Min*, **Nam Than***, Grace Hu, Woojung Shin, Yoko Ambrosini, Yong Cheol Shin, and Hyun Jung Kim. Probiotics-Mediated Epithelial Regeneration and Anti-Inflammatory Responses in a Pathomimetic "Leaky Gut Chip". *Digestive Disease Week (DDW)*, **2022**, San Diego, CA, USA (*Equally contributed; *Speaker; Podium presentation on May 22, 2022)
13. **Nam Than** and Melinda Simon. COMSOL Simulation and Analysis of the Effects of Gap Size Variation with a Constant Row Shift Fraction on Separation Resolution in Microfluidic Deterministic Lateral Displacement. *Biomedical Engineering Society (BMES) Annual Meeting*, **2020**, Virtual. (Poster)
14. **Nam Than**, Amy Bowers, Mohammed Abdi, Ron Labby, Andrea Fasoli, Alberto Mannari, Luisa Bozano, and Aminat Adebisi. An Electronic Nose to Differentiate Bacterial Species Based on Their Metabolic Volatile Organic Compounds. *The Society for Laboratory Automation and Screening (SLAS)*, **2020**, San Diego, CA, USA. (Poster)
15. **Nam Than**, Amy Bowers, Mohammed Abdi, Sarath Swaminathan, Ron Labby, Andrea Fasoli, Alberto Mannari, Luisa Bozano, and Aminat Adebisi. In vitro Predictive Model for Urine Infected with Escherichia coli. *American Society for Microbiology (ASM Microbe)*, **2019**, San Francisco, CA, USA (Poster)

PROCEEDINGS

1. Aminat Adebisi, **Nam Than**, Sarath Swaminathan, Mohammed Abdi, Amy N. Bowers, Andrea Fasoli, Alberto Mannari, and Luisa Bozano. [Rapid Strain Differentiation of E. coli-inoculated Urine Using Olfactory-based Smart Sensors](#). *Proceedings of the 5th International Conference on Sensors and Electronic Instrumentation Advances (SEIA)*, **2019**, 307.

INVENTION DISCLOSURES

Invention Disclosure Form

IDF 2025-176	Robust and Scalable Manufacturing of Ultrathin Porous PDMS Membranes , Cleveland Clinic Innovations, Cleveland, OH
IDF 2025-011	Multiplex Organ on a Chip (EXORM) , Cleveland Clinic Innovations, Cleveland, OH
IDF 2024-027	Design and creation of a Leaky Gut chip model for validating probiotic effects , Cleveland Clinic Innovations, Cleveland, OH
IDF 2022-170	A Multiplex Organ-on-a-chip , Cleveland Clinic Innovations, Cleveland, OH
Provisional Patent 26473.04465	Multiplex Organ on a Chip (EXORM) , Cleveland Clinic Innovations, Cleveland, OH

GRANT ACQUISITION ASSISTANCE

12/2024 – 12/2025	2024 Catalyst SPARK Award, Cleveland Clinic, Cleveland, OH <i>Principal Investigator: Hyun Jung Kim, PhD</i>
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RESEARCH SKILLS

Biomedical Engineering

Microfluidics

- Design & fabrication
- Photomask design; cleanroom fabrication; photolithography; soft lithography; deterministic lateral displacement (DLD) microfluidics; organ-on-a-chip design & fabrication; elastic & porous membrane fabrication
- Computational modeling
- Mathematical & computational fluid dynamics (CFD); finite element analysis (FEA); transport phenomena
- 3D Printing
- 3D modeling; DLP resin 3D printing; FDM filament 3D printing; print parameter optimization; support structure generation; multi-part assembly design; tolerance and fit optimization; post-processing; rapid prototyping

Organ-on-Chip

- Disease modeling
- Gut-on-a-chip; organ-on-a-chip design & fabrication; host-microbiome interaction; mechanical stimulation
- Organotypic biology
- Intestinal crypt isolation; intestinal organoid culture; mammalian cell culture; general cell line culture
- Microbiology
- Patient-derived fecal sample processing; anaerobic bacterial culture

Analytical & Data Interpretation

Biological Analytics

- Biochemistry
- DNA extraction; gel electrophoresis; ELISA; qPCR; permeability assays; protein and nucleic acid quantification
- Microscopy
- Brightfield imaging; phase-contrast imaging; DIC imaging; confocal fluorescence imaging; live-cell imaging; immunofluorescence microscopy; scanning electron microscopy; image analysis

Bioinformatics

- Transcriptomics
- Single-cell RNA-seq analysis; bulk RNA-seq analysis; differential gene expression analysis; pathway enrichment analysis (GSEA); cell clustering and annotation
- SNP analysis
- Identification of single base-pair variations in R
- Metagenomics
- 16S rRNA analysis; shotgun metagenomic analysis; microbial taxonomic profiling; microbial diversity analysis

Programming & presentation

Key Equipment | Software

- Spin coater; MA6 mask aligner; plasma cleaner; Corona treater | *OnShape; AutoCAD; SolidWorks; OpenSCAD*
- *COMSOL Multiphysics; MATLAB; Octave; Python*
- Asiga Pro4K; Fusion3 F200; Bambu Lab X2D; Anycubic Wash & Cure | *Asiga Composer; F3Slic3r; Bambu Studio*
- UVO Cleaner; FlexCell FX-6000; syringe pumps; peristaltic pumps
- Biosafety cabinet (Class II), CO₂ incubator; cell counter
- Anaerobic chamber; GasPak system; spectrophotometer

- GC-MS; NanoDrop; BioRad electrophoresis; plate reader
- Leica DMI8; Leica SP8 confocal; EVOS microscope | *LAS X; IMARIS; Fiji/ImageJ*
- *R packages: Seurat, DESeq2, clusterProfiler, fgsea; CellRanger*
- *R/Bioconductor*
- *R packages: DADA2, phyloseq; SILVA database*

- Programming
- HPC
- Visualization & presentation
- Data processing; automation scripting; artificial neural network (ANN)
- Linux command-line computing; shell scripting; SLURM job submission; remote cluster computing; workflow management
- Statistical analysis; data visualization; scientific illustration; publication-quality graphics; presentation design; scientific writing; manuscript preparation; poster design
- R; Python; Java; RStudio; Jupyter Notebook; PyCharm; Git/GitHub
- PuTTY; FileZilla; UNIX/Linux environment
- GraphPad Prism, Word; Excel; PowerPoint; Photoshop; Illustrator; BioRender; R/ggplot2

PEER REVIEW

- 06/2025 Advanced Biology, Wiley (*Mentor: Hyun Jung Kim, PhD*)
 05/2024 Biotechnology and Bioengineering, Wiley (*Mentor: Hyun Jung Kim, PhD*)
 10/2023 npj Biofilms and Microbiomes, Springer Nature (*Mentor: Hyun Jung Kim, PhD*)
 03/2023 Lab on a Chip, Royal Society of Chemistry (*Mentor: Hyun Jung Kim, PhD*)
 03/2022 Frontiers in Bioengineering and Biotechnology, Frontiers Media (*Mentor: Hyun Jung Kim, PhD*)

SERVICES & OUTREACH ACTIVITIES

- 01/2026 – Present **Member**, Communications Committee, Cleveland Clinic Research Trainee Association, Cleveland, OH
 08/2023 – 01/2026 **Co-chair**, Communications Committee, Cleveland Clinic Research Trainee Association, Cleveland, OH
 12/2023 – 12/2025 **Graduate Mentor**, Buddy Mentor Program, Cleveland Clinic Research Trainee Association, Cleveland, OH
 08/01/2024 **Judge**, Summer Student Poster Day 2024, Cleveland Clinic Research, Cleveland, OH
 09/2014 – 07/2015 **Patient Advocate**, Vietnamese Cancer Awareness Research and Education Society (VN CARES), University of California, Davis, Davis, CA
 03/22/2023 **Special Award Judge**, Northeastern Ohio Science and Engineering Fair, Cleveland, OH
 03/12/2020 (Virtual) **Category Judge**, Synopsys Championship Science Fair, San Jose, CA
 12/2013 – 06/2020 **Vice President**, Junior Legion of Mary, Our Lady of La Vang Catholic Church, San Jose, CA

MENTORSHIP

- Cleveland Clinic**, Cleveland, OH
- 01/2026 – Present Jose Torrado-Rivera *Current: PRIME Post-Baccalaureate Fellow (Case Western Reserve University)*
- 01/2025 – Present Amit Kumar *Current: Graduate Research Assistant (Cleveland State University)*
- 09/2024 – Present Zareh Dechkounian *Current: Undergrad research student (Case Western Reserve University)*
- 09/2024 – 04/2025 Joo Young (Sarah) Kim *Former: Undergrad research student (Case Western Reserve University)*
Current: Volunteer (UH Rainbow Babies & Children's Hospital)
- 09/2024 – 09/2025 Sohee Kim *Former: Undergrad research student (Cancer-focused Summer Undergraduate Research program; Case Western Reserve University)*
Current: Medical ICU Clinical Technician (Cleveland Clinic)
- 05/2024 – 08/2024 Meena Chandrasekharan *Former: Undergrad research student (Velosano Student Fellow)*
Current: Integrated Business and Engineering Honors Student (Biomedical Engineering, Ohio State University)
- 12/2023 – 07/2024 Mingzy (Cynthia) Zhang, MD; *Former: Research Fellow (Cleveland Clinic in Cleveland)*
Current: General Surgery Resident (Cleveland Clinic in Florida)
- The University of Texas at Austin**, Austin, TX
- 08/2021 – 07/2022 Grace Hu, MS *Former: Undergraduate Student (UT Austin)*
Current: Associate Scientist II (Abbvie)

MEMBERSHIPS

- 04/2022 – 04/2024 American Gastroenterological Association (*Student member*)
 01/2020 – 01/2024 Society for Laboratory Automation and Screening (*Student member*)
 04/2019 – 04/2021 The Honor Society of Phi Kappa Phi (*Inducted member*)
 01/2019 – 01/2021 American Society for Microbiology (*Student member*)
 01/2018 – 01/2024 Biomedical Engineering Society (*Student member*)

